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Dalti : A Queue Management System for Medical Clinics

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Dedication

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ملخص

في ظل التطور المتسارع للتكنولوجيا الرقمية، لا يزال القطاع الصحي الخاص في الجزائر يعتمد على أساليب تقليدية في إدارة طوابير الانتظار، كالقوائم الورقية والحجز عبر الهاتف. هذه الأساليب تُفضي إلى فوضى في الانتظار، وتلاعب في الأدوار، وغياب تام للتواصل بين العيادة والمريض.

تتمحور الأطروحة حول إيجاد حل لمشكلة إدارة طوابير الانتظار في العيادات الخاصة الجزائرية، من خلال تطوير منصة رقمية متكاملة تُعالج هذه الإشكاليات البنيوية وتُحسّن تجربة كل من المريض والعيادة.

يتضمن الحل المقترح تطوير منصة رقمية تحت اسم **Dalti**، تُتيح للمرضى حجز أدوارهم عن بُعد وتتبّعها لحظياً مع استقبال إشعارات فورية، كما تُزوّد العيادات بأدوات متقدمة لإدارة قوائم الانتظار وتطبيق قواعد الأولوية وإضافة المرضى الوافدين مباشرةً. المنصة مدعومة باللغة العربية, الفرنسية و الانجليزية وتعمل وفق نموذج Freemium يجعلها في متناول العيادات بمختلف أحجامها.

الكلمات المفتاحية: إدارة الطوابير، العيادات الخاصة، الجزائر، تطبيق جوّال، الحجز الرقمي،Freemium ، دالتي .

Abstract

Despite the rapid development of digital technology, the private healthcare sector in Algeria still relies on traditional methods for managing waiting lists, such as paper lists and telephone bookings. These methods lead to chaotic waiting times, manipulation of appointments, and a complete lack of communication between clinics and patients.

This study focuses on finding a solution to the problem of managing waiting lists in Algerian private clinics by developing an integrated digital platform that addresses these structural issues and improves the experience for both patients and clinics.

The proposed solution involves developing a digital platform called **Dalti**, which allows patients to book appointments remotely and track them in real time with instant notifications. It also provides clinics with advanced tools for managing waiting lists, implementing priority rules, and adding incoming patients directly. The platform supports Arabic, French, and English and operates on a freemium model, making it accessible to clinics of all sizes.

Keywords: waiting list management, private clinics, Algeria, mobile application, digital booking, Freemium, Dalti.

Résumé

Malgré le développement rapide des technologies numériques, le secteur de la santé privée en Algérie s'appuie encore sur des méthodes traditionnelles de gestion des listes d'attente, telles que les listes papier et les réservations téléphoniques. Ces méthodes engendrent des temps d'attente chaotiques, des manipulations des rendez-vous et un manque total de communication entre les cliniques et les patients.

Cette étude vise à trouver une solution au problème de la gestion des listes d'attente dans les cliniques privées algériennes en développant une plateforme numérique intégrée qui remédie à ces problèmes structurels et améliore l'expérience des patients et des cliniques.

La solution proposée consiste à développer une plateforme numérique appelée **Dalti**, qui permet aux patients de prendre rendez-vous à distance et de suivre leur prise de rendez-vous en temps réel grâce à des notifications instantanées. Elle fournit également aux cliniques des outils avancés pour la gestion des listes d'attente, la mise en œuvre de règles de priorité et l'ajout direct des nouveaux patients. La plateforme est disponible en arabe, en français et en anglais et fonctionne selon un modèle freemium, la rendant accessible aux cliniques de toutes tailles.

Mots-clés : gestion des listes d'attente, cliniques privées, Algérie, application mobile, réservation numérique, Freemium, Dalti.

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General Introduction

Project Background

In Algeria, private healthcare accounts for a large share of daily outpatient visits. Many people prefer private clinics and medical offices as an alternative to the public healthcare system. However, appointment booking is still largely conducted by phone, queues are often managed with paper lists, and walk-in patients are manually tracked by receptionists.

This situation does not reflect a lack of interest in digitalization. Rather, it highlights the absence of widely adopted digital solutions specifically designed for outpatient queue management in Algeria. Most healthcare digitalization initiatives, including hospital information systems, telemedicine platforms, and national e-health programs, have primarily focused on clinical data management and public healthcare services. Consequently, automated queue management has received limited attention in both the public and private sectors, leaving private practices to face persistent patient flow challenges without adequate digital support.

Problem

This gap causes ongoing problems. Paper queue lists can be easily changed, so patients might move or replace them to get ahead. Phone bookings do not guarantee that patients will be seen on time, since walk-ins and overbooking often push back scheduled appointments without notifying patients. If a doctor is absent or the clinic unexpectedly closes, patients are not informed and may travel far only to find out upon arrival.

Besides these problems, patients who get a spot in line often must wait in the waiting room for a long time, with no idea how long it will take or if the doctor is still available.

If the doctor must leave or cancel appointments, patients usually only find out after waiting a long time. This lack of clear information places all the responsibility on patients, leaving them without updates or options.

These problems happen every day. Because of this, both patients and clinic staff struggle to plan their time. This is more than just a minor inconvenience. The main issue is the lack of a reliable, secure, real-time system to manage patient queues in private outpatient clinics.

Proposed Solutions

This thesis introduces a digital queue management platform created for the specific needs of private clinics in Algeria. The platform is built to support two main user groups, each with different requirements.

Patients can reserve a spot in the queue remotely, without being at the clinic. Each booking uses a unique code to ensure a fair process. Patients can see their place in line in real time and get notifications when their turn is near. The platform supports both Arabic, French, and English and lets users book for family members, which is common in Algeria. The service for patients is always free.

Clinics get a full queue management system that can handle multiple queues simultaneously, link queues for visits with multiple steps, and add walk-in patients without disrupting the schedule. The system uses a first-in, first-out (FIFO) order but can also consider factors such as age or urgency when needed. Clinics can choose from four service levels, starting with a free trial for small practices and moving up to a paid Enterprise plan for larger clinics.

Document Plan

This thesis is organized into two chapters. **The first chapter** situates the project through a structured review of the current landscape, including the state of queue management systems in healthcare, the trajectory of digitalization in the Algerian health sector, an analysis of solutions in comparable contexts, and a synthesis that identifies the specific gap this platform addresses. **The second chapter** presents the complete contribution, detailing the system requirements, architecture, UML design, implementation, and financial analysis.

1

1 State of the Art

Introduction

This chapter places the current project in the wider field of digital queue management in healthcare. It starts by describing how private outpatient clinics in Algeria operate and the patient flow problems that led to this research. Next, it reviews related work in two areas: how healthcare queue management systems have developed globally and how healthcare digitalization has progressed in Algeria, especially focusing on local digital solutions. The chapter ends by summarizing the findings, pointing out gaps that current solutions do not address, and explaining where the proposed platform fits in the existing landscape.

1.1 Context and Problem Statement

In Algeria, private clinics and individual medical offices handle much of the daily outpatient care. Unlike public hospitals, which use formal administrative systems, these private clinics rely on manual processes managed by staff. Appointments are made by phone, queues are kept on paper, and walk-in patients are handled by receptionists without a standard system. For most private practices in Algeria, this is the usual way of working, not just a temporary situation.

The main problems with this system are:

- **Queue integrity**: Paper queue lists posted outside clinics are not secure. Any patient can remove or change the list to move themselves up.
- **Scheduling reliability**: Booking by phone does not guarantee on-time appointments. Overbooking and untracked walk-ins often push scheduled patients aside, and patients are not told about these changes.

- **Patient communication**: Patients are not told about delays, cancellations, or clinic closures. They only find out when they arrive at the clinic.
- **Staff support**: Receptionists handle these problems by hand and as they happen. They do not have tools to help reorder the queue, assign priorities, or communicate with patients.

These problems are not just occasional; they are built into a system that does not use digital tools to manage patient flow. While healthcare in Algeria is becoming increasingly digital through e-health projects, hospital information systems, and telemedicine, these efforts have largely ignored how outpatient queues are managed. Right now, there is no widely used solution for this problem in Algerian private clinics. This gap is the main reason for this research.

1.2 Queue Management Systems in Healthcare

1.2.1 Evolution Overview

Researchers have extensively documented patient waiting times in outpatient settings worldwide. Long lines at registration, unexpected delays, and no real-time updates often cause problems that lower patient satisfaction and make clinics less efficient [1, 2]. Traditional queue management depends on people being there in person, paper lists, and staff working by hand. As more patients come and care becomes more complex, these old methods are no longer sufficient. The table below summarizes the key stages in the evolution of digital queue management systems in healthcare, from early web-based registration to AI-driven optimization.

Table 1.1 : The evolution of queue management systems in Healthcare

Period	Technology	Key Advancement	Representative Work
Early2000s–2010s	Web-based registration	Patients schedule appointments remotely; reduces walk-in congestion but leaves real-time man unsolved	[3]

2010s–2020	Mobile applications	Live queue status and push notifications; patients can monitor their position and wait outside the clinic until called	[4, 5]
2015–2020	Priority-based scheduling	Weighted queue ordering by acuity, urgency, and wait duration; accumulating priority functions prevent indefinite delays for lower-priority patients	[6, 7]
2020–present	AI-driven optimization	Agentic AI and reinforcement learning applied to real-time urgency reclassification and dynamic patient reallocation across consultation slots	[8]

It is worth noting that these advances have been largely achieved in developed countries with well-established digital health infrastructures. The local context in Algeria remains far from this stage, where even basic digital tools for patient flow management are still absent in most outpatient settings.

Xijing Hospital was among the first to use a web-based appointment system, which helped cut wait times and improve patient satisfaction (**Cao et a**l. [3]). Later, mobile tools like **EasyHos** connected hospital databases to patients' devices, allowing real-time updates on queues and notifications (**Vorakulpipat et al.** [4]). More recently, an AI system for urgency mapping and queue management has been used to adjust patient priority, reduce service misallocation, and better spot critical cases (**Gupta et al.** [8]).

1.2.2 QR Code Integration

Another development brings QR code scanning into clinic operations. Systems like QKlinik [5] let patients register, reserve their place in line from afar, check wait times online, and handle pharmacy transactions on a single platform. User tests and black-box testing found that QKlinik scored an average usability rating of 4.78 out of 5 among patients, doctors, and clinic staff [5]. This method is especially useful in clinics with heavy workloads and few staff.

1.2.3 Identified Gap

Most current systems were built for hospital outpatient departments in well-resourced settings, where stable internet, trained staff, and integrated information systems are already in place. In contrast, small private clinics often have fewer staff, unreliable internet, walk-in patients, and no existing digital systems. These clinics have not received much attention, but this is exactly where the current platform is used.

1.3 Healthcare Digitalization in Algeria

1.3.1 Historical Trajectory

Algeria has a history of advancing digital healthcare, but results at the facility level remain limited. The first major step was E-Algeria 2013, a national plan to connect public health institutions with fiber-optic networks and set up electronic systems for managing patient data and appointments [9]. Later, more plans were made to push digital changes in public administration, including health services [9]. However, even after more than 10 years, most public hospitals are only partially digital, and patient reception and queue management still primarily use paper [10].

1.3.2 Conditions for Digital Adoption

Today, it is easier to move toward digital healthcare than it was ten years ago. Studies in Algeria show that people are increasingly willing to use digital health services, mainly because they find them useful and easy to use, and because they have positive views of technology. Electronic word of mouth also helps [11]. The COVID-19 pandemic accelerated this change: health authorities quickly rolled out telemedicine, and both patients and providers adapted to technology more quickly than expected [12]. People are not opposed to digital health tools; the main problems stem from infrastructure, institutional constraints, and how systems are designed.

1.3.3 Persistent Barriers

Even though people are ready to use digital health tools, there are still major barriers to making digitalization last:

- Unreliable internet connectivity, particularly in clinic settings outside major urban centers.
- Administrative inertia and insufficient staff training at the institutional level.
- Unresolved data privacy concerns and the absence of a coherent regulatory framework for digital health [9, 10, 12].

These constraints have directly informed the design requirements of the present platform, as detailed in Chapter 2.

1.4 Digital Solutions in the Algerian Health Sector

Digital tools have been progressively introduced across Algeria's public hospital network over the past few years. The following table summarizes the principal domains of intervention and their relevance to the present project.

Table 1.2 : Digital Solutions in the Algerian Health Sector

Domain	Scope	Relevance to This Project
Hospital Digitization	Electronic medical records, HR systems, pharmacy supply chain, examination and treatment tracking [9, 10]	Confirms that digitalization is advancing but patient flow and queue management remain untouched.
Telemedicine	Remote consultations, chronic disease management, maternal health programs, national portal development [11, 12]	Addresses geographic distance, not in-clinic congestion; a structurally different problem.
COVID-19 Response	Digital platforms for public health awareness and care continuity during lockdowns [10]	Normalized digital health interactions; lowered psychological barriers to adoption.

Interoperability	IHE-profile methodologies for data exchange between health institutions [13]	Defines a future integration pathway for this platform as the national ecosystem matures.

In contrast to these developments, outpatient queue management initiatives in private clinic settings remain notably absent. Yet, research applying an extended Technology Acceptance Model in the Algerian context confirms that citizens are willing to adopt digital health tools when they are perceived as useful and easy to use [11]. The primary barrier is not patient resistance; it is the absence of a solution designed for this specific context.

1.5 Key Dimensions of Healthcare Queue Management Systems

Queue management systems in healthcare can be understood through four intersecting dimensions. The table below provides this classification framework and positions the present platform within it.

Table 1.3 : The Classification of Queue Management Systems in Healthcare

Dimension	Variants	Representative Works	This Platform
Patient Presence	In-person token dispenser → Remote web/mobile reservation	[14, 15]	Remote booking via mobile app; no physical presence required. Walk-in patients handled via staff entry.
Queuing Logic	FCFS → Priority-based (age, urgency, visit type) → AI-driven dynamic reallocation	[16,17,18]	FIFO as default discipline; staff-controlled priority overrides; aging functions prevent indefinite deferral for lower-priority cases.

Deployment Model	On-premise standalone → Web/cloud → Mobile-first with push notifications	[14,19]	Cloud-hosted backend; Flutter mobile app for patients; Flutter Desktop dashboard for reception; push notifications via FCM/APNs.
Stakeholder Scope	Patient-only booking → Fully integrated patient + clinic administration	[15,16]	Integrated: patient intake, receptionist workflow, multi-queue configuration, and multi-stage routing within a single platform.

1.6 Queue Scheduling Algorithms in Healthcare Systems

Queue scheduling algorithms decide how patients are organized and called for service. Researchers have found three main approaches, each suited to a different clinical setting.

1.6.1 First-In, First-Out (FIFO)

FIFO serves patients in the order they arrive, regardless of condition or age. It is popular because it is easy to use and considered fair [1]. Web-based FIFO systems have improved resource use and patient satisfaction in outpatient clinics [20]. However, FIFO does not account for different levels of clinical urgency [21], which is mainly an issue in emergency care rather than in scheduled appointments.

1.6.2 Static Priority Queue

Priority scheduling assigns each patient a level based on clinical needs, urgency, age, or visit type, and always serves the highest-priority patient next. This method helps prevent the risky delays that FIFO can cause in high-acuity settings [22]. However, it can also mean that lower-priority patients wait much longer if urgent cases keep arriving [23].

1.6.3 Accumulating Priority Queue (APQ)

The APQ builds on static priority scheduling by adding waiting time to the priority score [24]. If a low-acuity patient waits too long, their priority increases enough that they will be seen before a new high-acuity patient, so they are not delayed forever. This reflects what often happens in practice: doctors are more likely to call a patient as their wait time approaches the clinic's target [25].

1.6.4 Comparative Overview

FIFO works best when patients have similar needs, and fairness is the main goal. Static priority scheduling is suitable for situations where patient urgency varies, and quick action is needed for critical cases. The APQ combines both methods, using waiting time to make sure lower-priority patients are not always left out. The best choice depends on the clinical setting.

1.7 Analysis of Existing Solutions

This analysis reviews seven platforms, starting with the most established global solutions and ending with the closest academic example. Each step brings us closer to the specific problem this project addresses. This order shows how each reference narrows the gap and highlights ongoing unmet needs in the target context.

1.7.1 European Scale — Doctolib

Doctolib is the leading scheduling management platform in the French-speaking European market, serving over 450,000 healthcare professionals and 80 million patients [26]. It offers calendar management, patient reminders, and real-time appointment changes, and is certified to ISO 27001 and ISO 27701 standards [26, 27].

However, it is designed only for fixed appointment scheduling, does not manage walk-in queues or real-time queue tracking, and is not available in Algeria.

Figure 1.1: Doctolib Logo

1.7.2 MENA Scale — Vezeeta

Vezeeta, founded in Egypt in 2012, is the top digital health platform in MENA. It has raised $63 million and expanded to Egypt, Saudi Arabia, Jordan, Lebanon, Kenya, and Nigeria [28]. It offers appointment booking, teleconsultation, and pharmacy delivery, with support for Arabic and English interfaces [29].

Despite its size and cultural closeness to Algeria, Vezeeta uses a fixed-appointment model and does not offer real-time queue tracking or walk-in management, nor does it operate in Algeria.

Figure 1.2: Vezeeta Logo

1.7.3 Regional Scale — DabaDoc

DabaDoc, founded in Morocco in 2014, now has over 12 million users and 10,000 healthcare professionals in more than 100 specialties [30]. It operates in Morocco, Algeria, Tunisia, Nigeria, the Ivory Coast, and South Africa [31]. Its support for Arabic and French and its presence in Algeria make it the most comparable solution in the target market.

However, like Doctolib and Vezeeta, DabaDoc uses an appointment-scheduling model and does not offer real-time queue tracking, walk-in management, or clinic-side queue administration tools [30].

Figure 1.3: DabaDoc Logo

1.7.4 Algerian Platforms

SihhaTech, launched by the Algerian Foundation for Innovation and Development (AFIND), is a local e-health platform that connects patients, healthcare providers, and pharmacies [32]. It supports online appointment booking, in-clinic visits, phone consultations, and home visits, and is available as a mobile app and web platform [32]. Its main value is its direct use in Algeria.

However, it does not offer real-time queue tracking, multi-queue setup, priority-based ordering, walk-in handling, or offline mode.

Figure 1.4: SihhaTech Logo

eSiha, founded in 2019,is an Algerian e-health platform that offers appointment booking, audiovisual teleconsultation, digital prescriptions, and real-time monitoring for patients with chronic conditions [33]. With over 50,000 downloads on the Play Store and coverage across all specialties and regions, it is the most ambitious local platform [33].

However, it focuses on healthcare access and clinical record management, not queue operations. Walk-in handling, real-time queue reordering, multi-queue management, and offline features are not included.

Figure 1.5: eSiha Logo

Touri is an Algerian app that lets patients find doctors, book appointments, and get remote turn alerts without waiting in the clinic. It won an international telemedicine prize in December 2019 [34]. It is the closest local example to the queue-awareness model used by this platform. However, user reviews mention ongoing reliability issues, with confirmed reservations not always shown in the queue system [34]. Multi-queue management, priority rules, walk-in handling, and offline features are not included.

Figure 1.6: Touri Logo

1.7.5 Academic Reference — QKlinik

QKlinik, developed by Fauzi and Suhaimi at Universiti Kebangsaan Malaysia, is a web-based queue management system. It lets patients reserve queue numbers via QR codes, track real-time wait status, and complete pharmacy transactions through a single interface [10]. It was tested with simulated user assessments and black-box testing, achieving an average usability rating of 4.78 out of 5 [35]. As the most technically similar academic reference for this project, QKlinik shows that integrated online queue reservation works in outpatient settings. Its main limits are scope: it was tested only in a simulated single-clinic environment and does not cover multi-queue management, mobile push notifications, a Freemium model, or intermittent connectivity [35].

1.7.6 Summary Comparison

The table below consolidates the feature comparison across all seven platforms and the present project.

Table 1.4 : Featurecomparisonmatrix:existing solutions vs. this platform

(Rows: platforms. Columns: Real-time Queue Tracking / Walk-in Management / Multi-queue / Multilingual Profile /Offline Mode / Guest Booking/ Algeria Presence / Primary Focus)

Platform	RTQ	WM	MQ	ML-P	Offline Mode	Guest Booking	Algeria	Primary Focus
Doctolib	✗	✗	✗	Partial	✗	✗	✗	Appointment scheduling
Vezeeta	✗	✗	✗	✓ AR/EN	✗	✗	✗	Appointment scheduling
DabaDoc	✗	✗	✗	✓ AR/FR	✗	✗	Partial	Provider discovery
SihhaTech	✗	✗	✗	Partial	✗	✗	✓	Appointment + telehealth
eSiha	✗	✗	✗	Partial	✗	✗	✓	Telehealth + EMR
Touri	Partial	✗	✗	Partial	✗	✗	✓	Remote queue alerts
QKlinik	✓	✓	✗	✗	✗	✗	✗	Queue mgmt (prototype)
This Project	✓	✓	✓	✓ **AR/FR /EN**	–	✓	✓	**Queue ops + clinic admin**

1.8 Synthesis and Discussion

This chapter's analysis shows a clear pattern in the current landscape. From this, three main gaps stand out:

- **Gap 1:** There is no integrated solution for both walk-in and remote queues. At present, no platform combines real-time remote queue reservations with walk-in patient management in a single system designed for small private practices.
- **Gap 2:** There is a lack of operational features for clinics in Algeria. No local solution provides multi-queue setup, priority rules, or walk-in management tools that reception staff need to reliably and handle daily patient flow.
- **Gap 3:** There is no package adapted to the local context. No solution covers all Algerian-specific needs, such as proxy booking for family, guest reservations without an account, a trilingual Arabic-French interface with language settings for each entity, and a freemium model for small practices.

The platform created in this project aims to fill these three gaps. It is not a copy of Doctolib or DabaDoc's appointment systems, and it does not aim to align with eSiha's broader healthcare goals. Instead, it focuses on **real-time queue management for private outpatient clinics in Algeria**, with features for operational depth, local adaptation, and accessibility that current solutions do not provide.

Conclusion

This chapter lays the groundwork for the project by first describing how private outpatient clinics operate in Algeria. It then follows the development of digital queue management systems, starting with early web-based registration tools and moving to priority-based scheduling and AI-powered patient flow solutions. The chapter reviews the progress of healthcare digitalization in Algeria, noting both the advances made and the ongoing structural challenges. It also looks at seven platforms from Europe, the MENA region, and Algeria, and introduces a classification framework based on four key factors: patient presence, queuing logic, deployment model, and stakeholder scope.

A clear pattern appears digital health investment in Algeria has mainly focused on clinical data management and remote care delivery. However, the everyday problem of managing patient flow in private outpatient clinics has not been addressed. These findings set the stage for the contribution discussed in the next chapter

2 Contributions

Introduction

This chapter introduces the project's main contribution: the design and development of a digital queue management platform for private outpatient clinics in Algeria. It begins with an overview of the solution and its main components. Then, it outlines the system requirements, both functional and non-functional, based on the problem analysis from the previous chapter. The chapter also explains the system architecture and includes a UML design that shows the main workflows and how the components interact. It goes on to describe the development method used throughout the project. Finally, the chapter summarizes the technology stack and presents the final interface designs for both the patient mobile app and the clinic web dashboard.

2.1 Development Methodology

2.1.1 Methodology Selection

The project used the Agile methodology because it can handle changing requirements and allows for parallel development of the platform's two user-facing parts. Agile was chosen over plan-driven methods like Waterfall because the platform serves two distinct user groups with unique workflows, and requirements kept changing as real clinic usage was studied. A linear model would not have supported the frequent design changes needed during development.

2.1.2 Agile Framework Applied

The project was organized into eight-week development sprints. At the start of each sprint, user stories were chosen from the product backlog and prioritized by importance and estimated effort. Each user story described a specific feature from the viewpoint of a patient or clinic staff member, such as: "as a patient, I want to receive a push notification when my turn is approaching so that I do not need to wait inside the clinic." After each sprint, completed features were checked against acceptance criteria, and the backlog was updated with any new requirements found during development.

The product backlog was organized around two parallel tracks:

- **Patient track:** remote queue reservation, real-time position tracking, push notification delivery, proxy booking for family members, and a Trilingual interface.
- **Clinic track:** queue configuration, multi-queue management, priority rule setup, walk-in patient handling.

Dependencies between the two tracks, especially those involving queue-state synchronization for real-time tracking, were identified early. These were scheduled so that shared backend components would be ready before developing features that depended on them.

2.2 System Requirements

This section lists the system requirements for the proposed platform. It first defines the functional requirements, which describe the specific features the system must offer each user group. Then it outlines the non-functional requirements that set the quality and performance standards for these features.

2.2.1 Functional Requirements

A) Patient-Facing Requirements

- **Account Creation:** Register as a user via email with OTP (One-Time Password) verification, or proceed as a guest without creating an account.
- **Clinic Search:** Filter by name, medical specialty, or location.
- **Clinic Profile:** View working hours, active queues, and real-time wait status.
- **Remote Reservation:** Join a queue digitally without being physically present.
- **Secret Code:** Receive a unique secret code upon reservation for position tracking and check-in verification.

- **Live Tracking:** Monitor queue progress in real time with automatic updates.
- **Push Notifications:** Alerts are sent via FCM (Firebase Cloud Messaging) on Android and APNs (Apple Push Notification service) on iOS when the patient's turn is approaching, even if the app is closed. Both services require an active internet connection. If the connection is weak, SMS notifications can be used as a backup.
- **Cancellation & History:** Manage active bookings and review past visits.
- **Trilingual Interface:** Full support for Arabic, French, and English both at the account level and at the per-entity level (clinic names, queue names, and medical specialties displayed in the user's chosen language).
-
- **Free Access:** All patient features are provided at no cost.

B) Clinic-Facing Requirements

- **Organization Registration:** Set up the organization profile (clinic, medical center, or specialized practice), define operational hours, and configure the primary language and sector type.
- **Queue Configuration:** Create multiple or sequential queues (e.g., per physician or per treatment stage).
- **Automatic Linking:** Move patients between sequential queues without manual intervention.
- **Call Next Patient:** Trigger the next patient arrival from the reception interface.
- **Absence Recording:** Mark "no-shows" to automatically advance the queue.
- **Walk-in Management:** Add unregistered patients directly from the reception interface.
- **Priority Rules: Configure** priority rules layered on top of the base FIFO order, taking into account factors such as age, urgency, or visit type.
- **Shared Queues: Allow** patients to join either a shared queue across multiple doctors or a personalized queue linked to a specific physician, based on their preference or the next available appointment.

- **Receptionist Accounts:** Multi-user management with specific permission levels.
- **Service Points:** Configure service points within the facility, physical counters or consultation rooms, each assignable to a specific organization member. Service point shifts are tracked, recording the staff member who opened and closed each shift, along with the corresponding timestamps.

C)System-Level Requirements

- **Booking Code Integrity:** Ensure codes are unique, non-guessable, and validated.
- **Concurrency:** Maintain queue consistency and notification speed during high-traffic periods across multiple clinics.

2.2.2 Non-Functional Requirements

Table 2.1: Non-Functional Requirements

Quality Attribute	Requirements	Metrics
Performance	Real-time queue position updates delivered to patient devices. Push notifications are dispatched after the queue event is triggered. The clinic dashboard is responsive under concurrent receptionist operations.	/
Reliability	Clinic-side queue operations continue without interruption during connectivity failures. Server-side components maintain high availability.	/

Security	Beyond standard transport-layer protection, sensitive payloads — including authentication tokens and booking codes — are encrypted at the application layer before transmission. Patient identity and booking records are protected at rest. JWT tokens are scoped and time-limited.	Encrypted storage. Token-based auth with defined expiry. No cross-clinic data access is possible at the data layer.
Scalability	Architecture supports growth in registered clinics and concurrent active queues without structural changes to the codebase. Individual modules can be extracted into independent services if demand requires it.	Modular Monolith design with enforced internal boundaries.
Language and Locale Support	The full patient interface must be available in Arabic, Frensh and English. The Arabic interface must render correctly in RTL(Right to left) layout. Language switching takes effect immediately without requiring an application restart.	No application restart required for language switch.

2.3 Adopted Scheduling Algorithm

The platform uses FIFO as its primary scheduling method across all queue types. This method works well for general outpatient visits in private Algerian clinics, where patients book appointments with a specific doctor and urgent triage is not needed.

The main challenge here is not urgent cases, but stopping people from changing the queue order. Paper lists can be altered, and phone bookings lack timestamps, so it is hard to verify the order if someone complains. A digital FIFO system uses time-stamped bookings that can be tracked, helping prevent the manipulation that can occur with paper systems.

By default, the platform does not use automatic priority rules. Instead, staff can adjust the queue by hand. Authorized receptionists can change the order, move a patient forward, or add a walk-in, and every change is saved in an audit log. The system also allows clinics to set up optional priority rules, such as for patient age or urgency, if administrators want to use them.

This design keeps FIFO as the default to make the process transparent and fair, while also giving clinics the flexibility they need. It avoids setting up priority rules that might lead to unfair treatment.

2.4 System Architecture

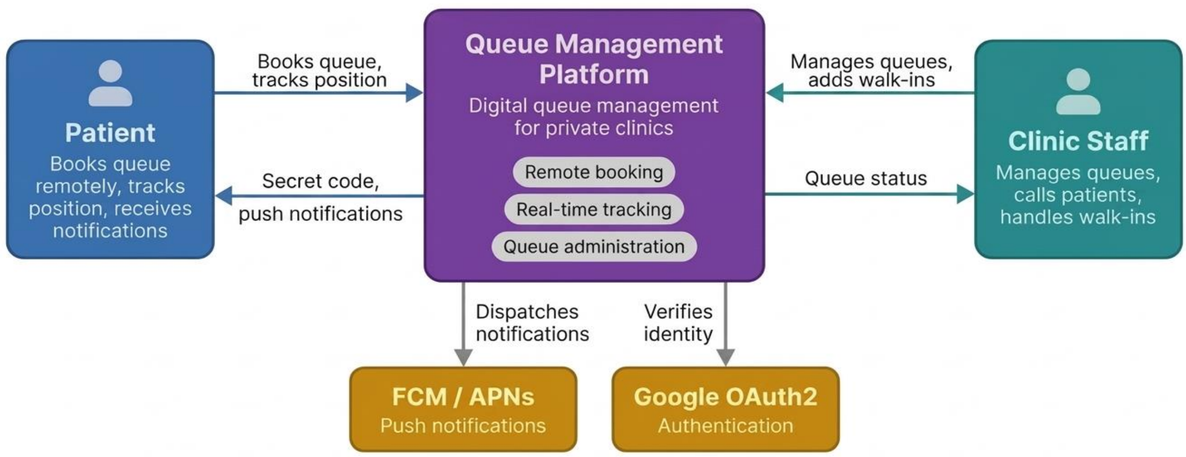


Figure 2.1: System Context Overview

Figure 2.1 shows the platform's system context with three main actors: the Patient, the Queue Management Platform, and the Clinic Staff. Patients use the platform remotely to book a queue spot, receive a secret code, and receive push notifications via FCM or APNs. Clinic staff use the platform to manage queues, call patients, and handle walk-ins. Authentication is handled through Google OAuth2. The platform's main features, remote booking, real-time tracking, and queue administration, are combined into a single deployable unit.

2.4.1 Architectural Overview

The platform uses a client-server setup. Three different client apps connect to a central backend using a RESTful API.

- **Flutter Mobile App:** This app is for patients and works on both Android and iOS.
- **Svelte Web Application:** This is a patient-facing interface for users who do not have the mobile app.
- **Desktop Dashboard:** This version is for clinics and is used by administrators and reception staff.

The backend uses a **Modular Monolith** architecture. It is a single deployable unit organized into five clear, loosely connected modules. The reasons for this choice are shown in the table below.

Table 2.2 : The Rationale for Choosing the Modular Monolith

Option	Advantage	Disadvantage	Verdict
Microservices	Independent scaling per service	Disproportionate operational complexity at the current scale	✗ Rejected
Unstructured Monolith	Simple to deploy	Accumulates tight coupling; difficult to scale later	✗ Rejected
Modular Monolith	**Simple deployment + enforced internal boundaries**	**Requires discipline in module isolation**	**✓ Selected**

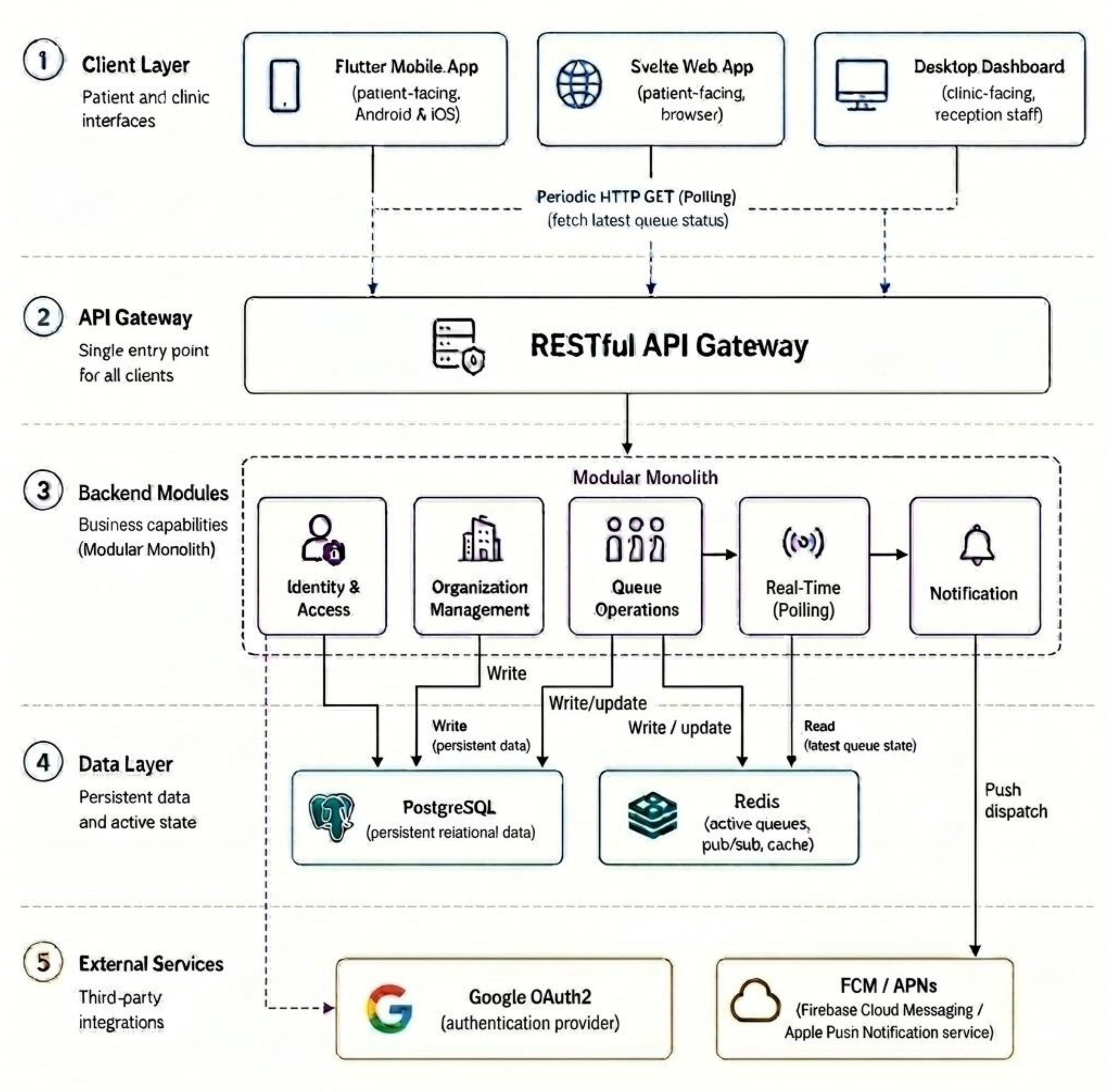


Figure 2.2: System Architecture Diagram

Figure 2.2 shows the system architecture, which is organized into five layers. The Client Layer has three interfaces: a Flutter Mobile App for patients, a Svelte Web App for browser access, and a Desktop Dashboard for clinic staff. Each interface connects to the backend by sending regular HTTP GET requests to get the latest queue state. These requests go through a RESTful API Gateway, which acts as the main entry point for all client interactions. The Backend Modules use a Modular Monolith pattern and are split into five main modules: Identity & Access handles authentication and authorization, Organization Management manages clinic profiles and queue settings, Queue Operations covers the full reservation and calling process, Real-Time distributes queue state updates using polling, and Notification sends push alerts to patient devices. The Data Layer uses PostgreSQL for storing relational data and Redis for managing the active queue state and broadcasting updates. The External Services layer uses Google OAuth2 for authentication and FCM/APNs to send push notifications across platforms.

2.4.2 Data Management

Two data stores are used, each assigned a distinct role:

Table 2.3 : The Two Data Stores

Store	Type	Role
PostgreSQL	Relational—persistent	Organization profiles and translations, user accounts, guest sessions, booking records (with secret codes), queue configurations, service points and shifts, and subscription data.
Redis	In-memory	Current state of active queues; publish-subscribe mechanism for broadcasting queue state changes to connected clients.

Real-time queue position updates are currently delivered to patient devices via polling. Redis manages the state changes that drive these updates.

2.4.3 Backend Module Structure

The relationships between the six backend modules and their connections to the data layer are illustrated in Figure2.3

Table 2.4 : Backend Module Structure

Module	Responsibilities
Identity & Access	Authentication, authorization, user account creation, token issuance and validation, and permission scoping. Organization member accounts are bound to their facility at the data layer — no cross-organization data access is possible. Guest session tokens are supported for reservations that do not require account creation.
Organization Management	Organization profile configuration (name, sector, location, and multilingual translations), working hours,queue creation and configuration, priority rule management, service point setup and shift tracking, organization member management with specialty assignment, and enforcement of Freemium subscription tiers.
Queue Operations	Lifecycle of every queue event: reservation, check-in validation, patient calling, absence recording, walk-in insertion, reordering, priority recalculation, and automatic inter-queue patient transfer. Maintains consistency between Redis state and PostgreSQL records.
Real-Time	Manages the distribution of queue state updates to connected client sessions. In the current implementation, updates are delivered via polling.
Notification	Receives trigger events from Queue Operations when a patient's position reaches the configured proximity threshold. Dispatches push notifications via FCM (Android) or APNs (iOS).

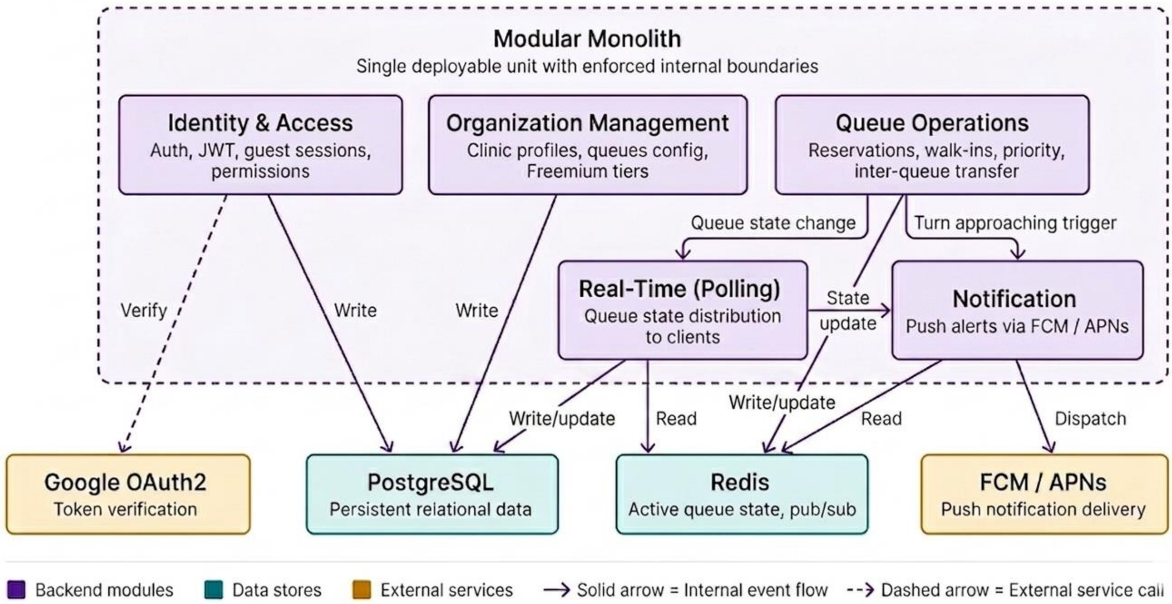


Figure 2.3: Backend Module Structure and Inter-Module Communication

Figure 2.3 illustrates the backend structure and how its modules communicate. The Identity & Access module manages authentication, issues JWTs, and controls permissions using Google OAuth2. Organization Management is responsible for clinic profiles, queue settings, and Freemium tiers, storing all data in PostgreSQL. Queue Operations manages reservations, walk-ins, priority rules, and transfers between queues. When a queue changes, the Real-Time module reads from Redis and sends updates to clients via polling. As a patient's turn approaches, the Notification module sends push alerts using FCM or APNs. Solid arrows show internal event flows, while dashed arrows show calls to external services.

2.5 UML Design

2.5.1 Use Case Diagram

The use case diagram identifies the three actors interacting with the platform and the operations available to each.

User (Patient)

- Registers an account via email, or proceeds as a guest without account creation
- Searches for organizations by name, specialty, or location
- Views an organization profile including working hours and active queues
- Makes a queue reservation remotely, receiving a unique secret code upon confirmation
- Views the current status and position of an existing reservation via secret code
- Cancels a reservation before being called
- Receives push notifications as their turn approaches

Organization Admin (Clinic)

- Authenticates and maintains an active session via token refresh
- Creates and manages the organization profile
- Manages organization subtypes and medical specialties
- Manages clinic members and their assigned specialties
- Creates and edits queues, including linking queues for multi-stage visits
- Manages service points and assigns them to queues
- Assigns queue managers to specific queues
- Configures priority rules based on age, urgency, or visit type

Queue Manager (Reception Staff)

- Authenticates and maintains an active session via token refresh
- Calls the next patient in the queue, resolving and advancing the active reservation record
- Postpones or reschedules a reservation within the current session
- Records patient absences to automatically advance the queue
- Adds walk-in patients directly from the reception interface
- Links queues for sequential patient routing

System Relationships

- **Cancel Reservation** includes **Authentication**, as cancellation requires identity verification
- **Register Account** includes **Authentication**, as account creation initiates the authentication flow
- All Receptionist and Admin operations include **Authentication,** as staff actions require an active verified session

Figure 2.4: Use Case Diagram

2.5.2 Flowchart (Queue Operations Flow) :

This flowchart depicts the complete lifecycle of patient queue interactions within the platform, encompassing three parallel entry paths: patient-initiated remote booking and manual walk-in entry by reception staff. After a patient joins the queue, either remotely using a unique secret code or as a walk-in added by the receptionist, the system manages the active queue by sequentially calling patients, recording absences, and automatically advancing the queue when a patient does not respond. Upon completion of service, the system determines whether the visit requires multiple stages.

If so, the patient is automatically transferred to the next linked queue, the queue position is closed, and the visit concludes.

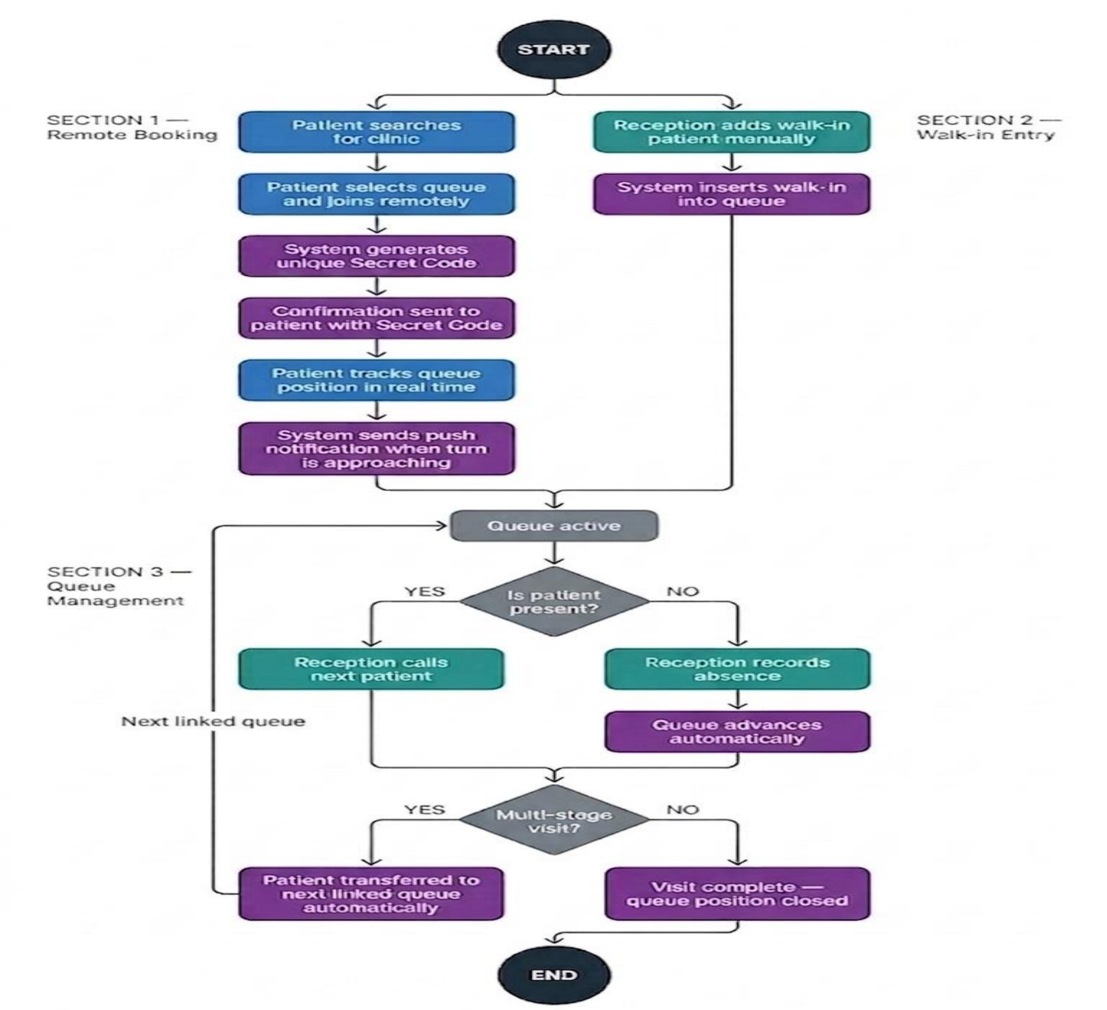


Figure 2.5: Queue Operation Flow

2.5.3 Class Diagram

The class diagram presents the full domain model of the platform, organized around five functional clusters.

Identity and Access

- **User**is the central identity record for all authenticated actors: patients, clinic staff, and queue managers
- **RefreshToken**supports stateless session renewal, with each user potentially holding multiple active tokens

Organizational Structure

- **Organization**represents a registered clinic or facility, storing geographic coordinates (Latitude, Longitude, Geom) for location-based search and a Sector attribute distinguishing public from private facilities
- **OrgTranslation** externalizes all user-visible strings — clinic name and address — keyed by language code, enabling independent Arabic and French translations without structural duplication
- **OrgSubtype** and **OrgSubtypeTranslation**classify facilities by type (general practice, dental clinic, specialist office), with full translation support

Specialties and Membership

- **Specialty**and **SpecialtyTranslation**model medical specialties with bilingual support, linked to organizations via **OrgSpecialty**
- **OrgMember**associates a User with an Organization and optionally with a specific Specialty; the IsPublic flag controls visibility in patient-facing search results

Queue Configuration

- **Queue**represents a single waiting queue within a facility
- The self-referential **NextQueueId**foreign key implements automatic queue linking: upon service completion, the system reads this field to forward the patient to the next stage without manual staff intervention

- **QueueTranslation**carries bilingual display metadata for each queue
- **QueueManager**tracks which staff members are authorized to manage a given queue
- **Reservation**captures a single patient queue position
- **SecretCode**holds the unique alphanumeric code generated at booking, used at check-in to verify the reservation and prevent queue manipulation
- **UserId**is optional: when present, the reservation is linked to an authenticated account; when absent, the Info field stores basic contact details (name and phone number) for walk-in or proxy-booked entries
- **ScheduledDate**and ScheduledTime support date-specific bookings where applicable
- **CalledByUserId**and **CalledAt**record, which staff member called the patient and at what time, contributing to the audit trail
- **NextQueueReservation**links an original reservation to its successor in the next queue stage, preserving multi-stage visit continuity
- **ReservationPerDay**maintains a running daily count per queue, supporting capacity monitoring without requiring aggregate queries at runtime

Service Points and Shifts

- **ServicePoint**represents a physical consultation window or desk within a facility, associated with an organization and optionally with a specific **OrgMember**
- **QueueServicePoint**links queues to the service points that serve them, enabling a shared-queue model where multiple physicians draw from a single queue
- **ServicePointShift**records operational sessions per service point: who opened the shift (**ManagerUserId**) and whether and by whom it was closed (**ClosedByUserId**, **ClosedAt**), providing temporal boundaries for per-session audit reporting

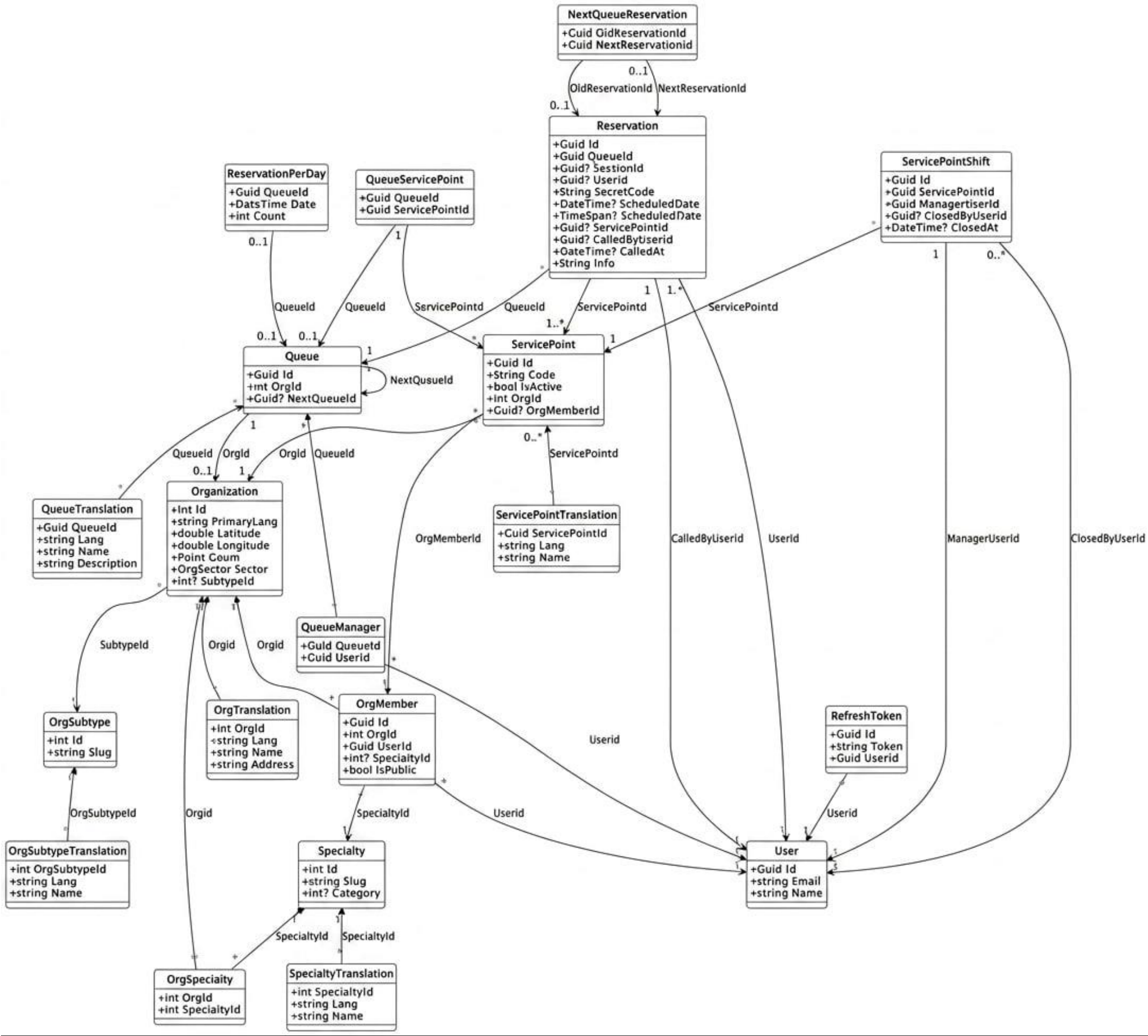


Figure 2.6 : Class Diagram

2.5.4 Sequence Diagram (Authentication flow)

This sequence diagram illustrates two authentication paths supported by the platform. In Flow A, an authenticated user signs in using Google OAuth2. The client application redirects the user to Google, receives a Google ID token, and forwards it to the backend through the API Gateway. The Identity & Access module then validates the token with Google and issues a scoped, time-limited platform JWT, which is used for all subsequent API requests. In Flow B, a guest user bypasses account creation. The client sends a POST request to /auth/guest, prompting the Identity & Access module to generate a temporary guest session token. This token grants the user limited access, sufficient to make a queue reservation without registration.

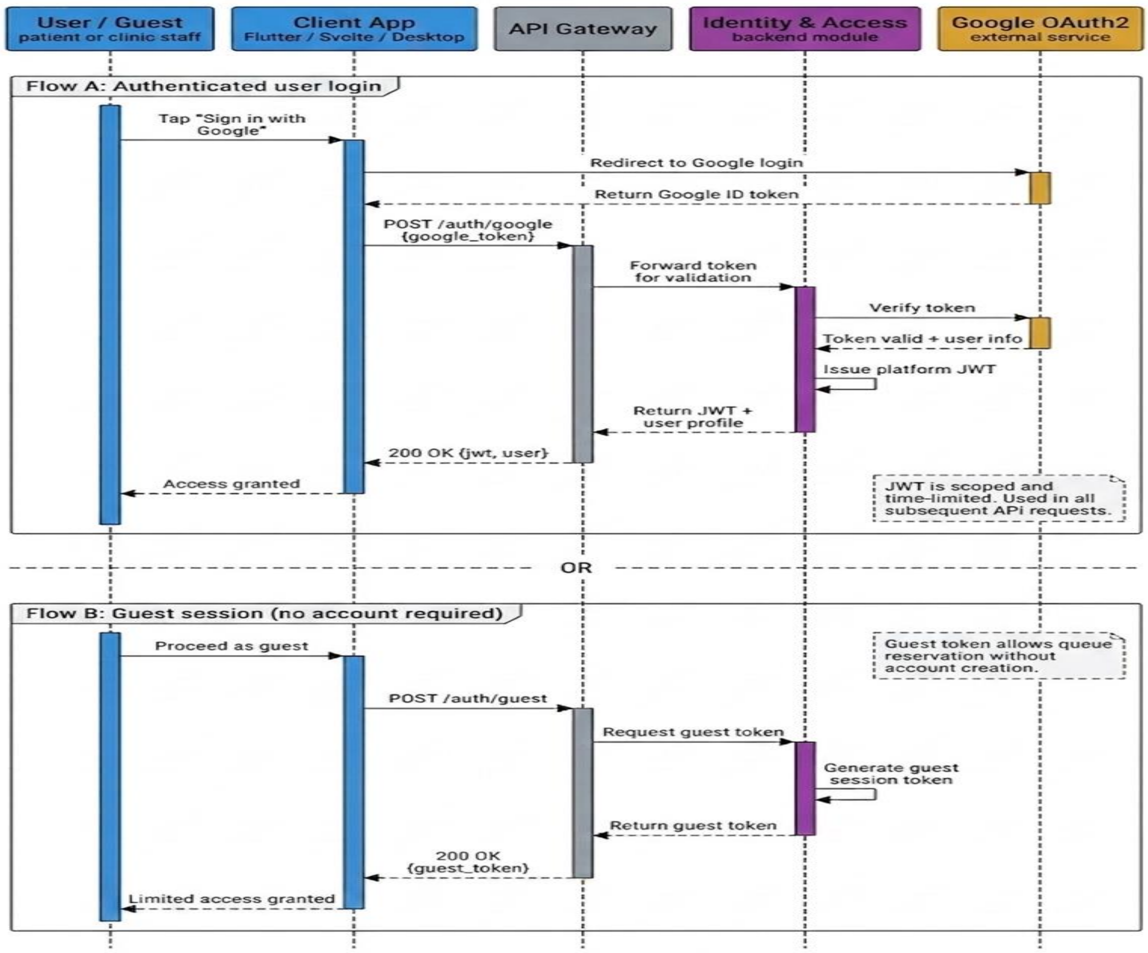


Figure 2.7: Authentication Flow

2.6 Technology Stack

The platform includes three main parts: a mobile app for patients, a desktop app for clinic administration, and a backend API. Each uses technologies chosen to meet the platform's needs and work well in Algeria.

2.6.1 Backend Technologies

Table 2.5 : Backend Technologies

Logo	Technology	Role & Justification
	.Net Core	High-performance, strongly-typed runtime chosen for its async I/O model and thread management, handling concurrent queue operations across multiple clinics.
	PostgresSQL	Relational database chosen for its integrity constraints, transactional reservation operations, and PostGIS geographic queries enabling location-based clinic search.
	Redis	In-memory state layer chosen for sub-millisecond queue read latency and pub/sub broadcasting, serving high-frequency patient polling without overloading the relational database.
	Google OAuth2 + JWT	Delegates credential management entirely to Google, eliminating password storage and brute-force risks; JWT tokens carry role and organization binding for all API requests.
	FCM / APNs	The only mechanism by which the operating system delivers system-level notifications to backgrounded apps is chosen because in-app polling alone cannot alert patients when their turn is approaching.

2.6.2 Frontend Technologies

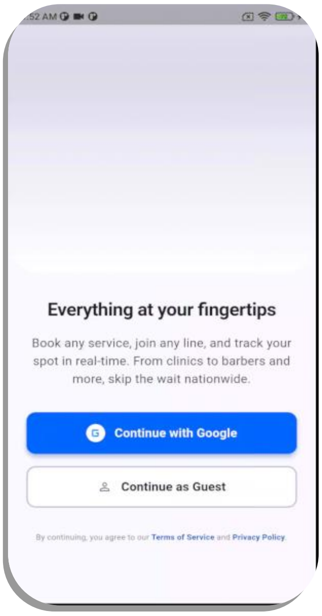
Table 2.6 : Frontend Technologies

Logo	Technology	Role & Justification
	Flutter	Cross-platform framework powering both the patient mobile app (Android/iOS) and the clinic desktop dashboard, chosen for native rendering performance and built-in RTL (Right to Left) Arabic support.
	Svelte	Compiler-based framework for the patient-facing web interface, providing access to users without a mobile app, with fast load times and minimal bundle size.

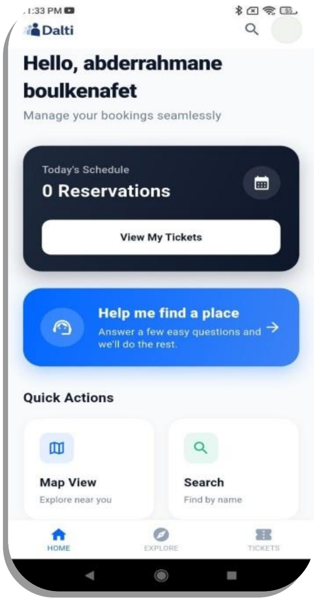
2.7 User Interfaces

2.7.1 Client Interfaces (Mobile App)

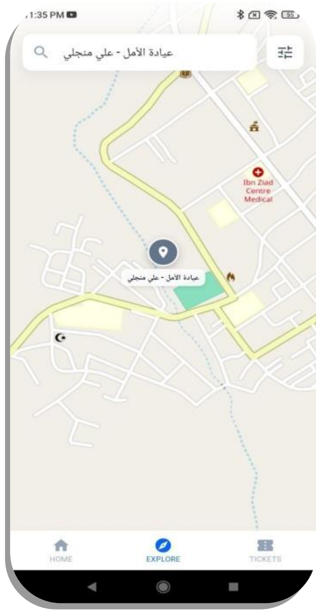
The patient-facing mobile application provides a streamlined reservation workflow accessible through two authentication paths: sign-in via Google OAuth2, or guest access without account creation (Screen 1). Upon authentication, the home screen presents the user's active reservations alongside quick navigation actions (Screen 2). The application integrates a map-based clinic discovery interface, allowing patients to locate and select nearby facilities (Screen 3), filter by medical specialization (Screen 4), and choose the appropriate queue (Screen 5). The reservation form captures the required patient details prior to confirmation (Screen 6). Upon successful submission, the system generates a unique secret code tied to the booking, serving as the patient's verification token (Screen 7). The digital ticket view displays this code alongside the patient's current position in the queue, providing real-time visibility into wait status (Screen 8).


Everything at your fingertips
Book any service, join any line, and track your spot in real-time. From clinics to barbers and more, skip the wait nationwide.
Continue with Google
Continue as Guest

1


Dalti
Hello, abderrahmane boulkenafet
Manage your bookings seamlessly
Today's Schedule
0 Reservations
View My Tickets
Help me find a place
Answer a few easy questions and we'll do the rest.
Quick Actions
Map View
Explore near you
Search
Find by name
HOME
EXPLORE
TICKETS

2


عيادة الأمل - علي منجلي
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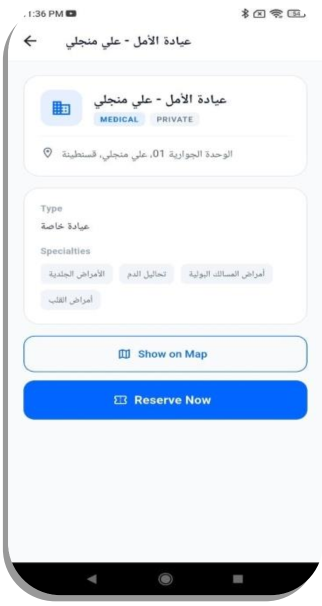
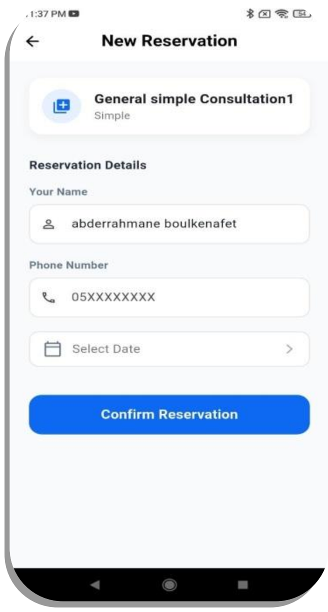

عيادة الأمل - علي منجلي
MEDICAL
PRIVATE
Type
عيادة خاصة
Specialties
Show on Map
Reserve Now

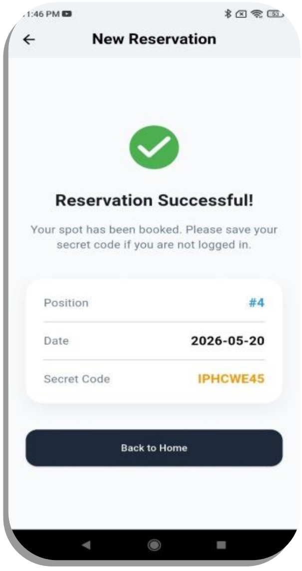
4


Select a Queue
General simple Consultation1
General medical checkup queue
No reservations today
Open
Mode: Simple
Proceed
طابور 2
0 slots taken today
Open
Mode: TimeBased
Proceed

5


New Reservation
General simple Consultation1
Simple
Reservation Details
Your Name
abderrahmane boulkenafet
Phone Number
05XXXXXXXX
Select Date
Confirm Reservation

6



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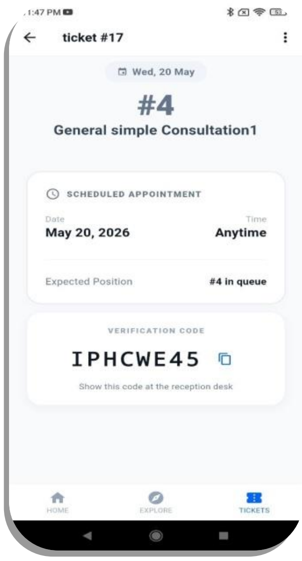

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Figure 2.8: Clients Interfaces (Mobile App)

2.7.2 Admin Interfaces (Desktop App)

Figure 2.9 illustrates the administration and operational interfaces of a clinic system. (screen 1) shows the institution's dashboard, which includes its details, medical specialties, and active waiting list settings. (screen 2) shows the live waiting list management interface for general reception, displaying real-time patient tracking, ticket status, and operational controls for calling the next patient.

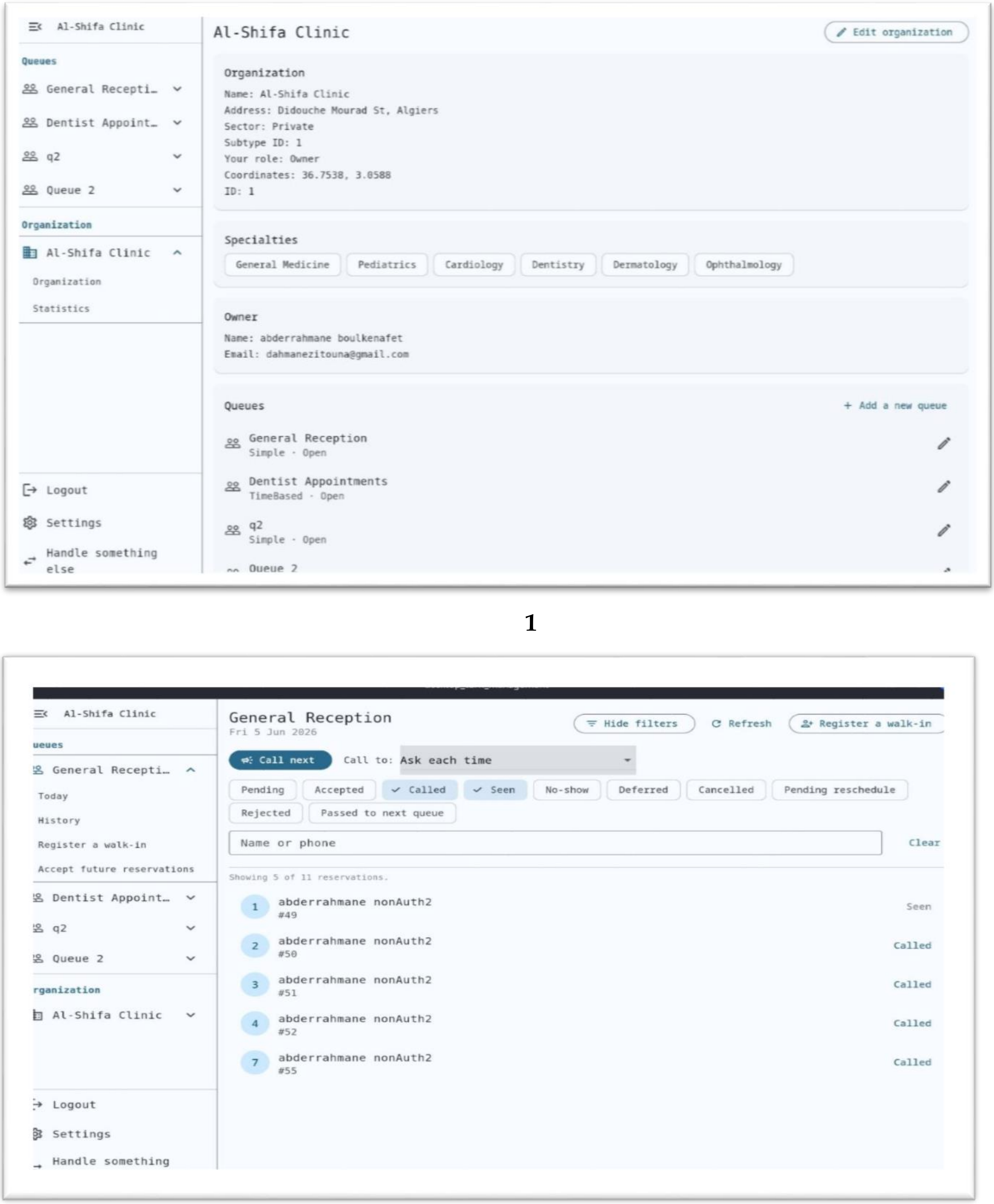

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Figure 2.9: Admin Interfaces

Conclusion

This chapter describes the main contribution: a digital queue management platform for private outpatient clinics in Algeria. The development process followed the Agile methodology. Features for patients and clinic staff were added and tested in separate phases.

At first, the chapter explained the functional and non-functional requirements based on the latest research. These requirements included features such as remote queue reservation, real-time tracking, support for multiple languages, walk-in management, handling multiple queues, and clinic administration.

Next, the chapter presented the system architecture. The client-server setup uses a modular monolith backend, which simplifies deployment and supports maintenance and future growth. Different modules handle identity management, organization administration, queue operations, real-time communication, and notifications. PostgreSQL and Redis are used to reliably store data and efficiently manage queue states.

The UML design showed the system's structure and behavior using use cases, flowcharts, class diagrams, and sequence diagrams. This gave a clear picture of how the platform works. The technology stack was chosen to meet performance, security, scalability, and multi-language requirements. The interfaces built demonstrated how design choices affect the experience for both patients and clinic administrators. The final platform is ready for use in Algerian clinics, featuring remote reservations, real-time tracking, useful tools, and adaptations to local needs. This system addresses current gaps in existing solutions and lays the foundation for future improvements.

General Conclusion

Synthesis

This thesis looked at how queues are managed in private outpatient clinics in Algeria. Most clinics still use paper waiting lists, phone reservations, and manual coordination by reception staff. These methods can lead to queue manipulation, a lack of transparency, poor communication, and long patient waiting times.

To address these problems, a digital queue management platform was created for Algerian private clinics. Patients can use it to book their place remotely, see real-time updates on the queue, get notifications when their turn is near, and use the service in Arabic, French or English. Clinics get tools for managing queues, handling walk-in patients, setting up multiple queues, managing service points, and controlling staff access.

Technically, the system uses a client-server setup with a modular monolith backend. PostgreSQL stores the data, and Redis handles real-time updates to the queue. The user interfaces are built with Flutter and Svelte for different use cases. UML modeling, careful design, and Agile development helped keep the project organized.

The platform provides a practical solution that makes processes more transparent, reduces administrative work, and improves the experience for both patients and clinic staff. By focusing on outpatient queues, this project fills a gap that current healthcare platforms in Algeria have not solved.

Perspectives

Although the current implementation meets the main requirements identified during the analysis, several features are planned for future development.

Right now, patient devices get queue updates through polling. In the future, WebSocket-based communication will be added to provide instant updates and lower both network and server load.

Offline Mode with Automatic Synchronization

Currently, clinics need a constant internet connection to use the system. An offline mode is planned, using IndexedDB to store data locally, so queue operations can continue during network outages and sync automatically when the connection returns.

Audit Log Module

A full audit logging system is planned to track all queue actions, including what was done, who performed the action, what it affected, and when it occurred. This will improve traceability, accountability, and oversight. It will also stop patients from making more than one active reservation at the same clinic on the same day, helping keep the queue fair and consistent.

Advanced Analytics and Reporting

Future versions may add dashboards showing key metrics such as average wait times, queue throughput, no-show rates, and service utilization. These tools will help clinics make data-driven decisions.

In summary, this platform lays a strong foundation for digital outpatient queue management in Algeria and provides a scalable system that can support future improvements.

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Acronyms

UML	Unified Modeling Language
OTP	One-Time Password
FCM	Firebase Cloud Messaging
APNs	Apple Push Notification service
RTL	Right-to-Left
JWT	JSON Web Token
APT	Application Programming Interface
EMR	Electronic Medical Record